

习 题 12.2

1. 设 $y=f(x)$ 是以 2π 为周期的函数, 它在 $[-\pi, \pi)$ 中的表达式分别由下列各式给出, 求出 $f(x)$ 的傅里叶级数及其和函数.

(1) $f(x)=x, -\pi \leq x < \pi;$ (2) $f(x)=x^2, -\pi \leq x \leq \pi;$

1. $f(x)$ 为奇函数, $a_n=0$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} x \cdot \sin nx \cdot dx = -\frac{2}{n\pi} \int_0^{\pi} x d\cos nx = -\frac{2}{n\pi} (x \cos nx \Big|_0^{\pi} - \int_0^{\pi} \cos nx dx) \\ &= \frac{-2}{n\pi} \cdot \pi \cdot (-1)^n \\ &= \frac{2}{n} (-1)^{n+1} \end{aligned}$$

$$\therefore f(x) \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$$

而 $f(x)$ 在 $(-\pi, \pi)$ 上分片连续且单调

$$\therefore f(x) = \begin{cases} 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx \\ 0, x = \pm \pi. \end{cases}$$

2. $f(x)$ 为偶函数

$$b_n=0, \quad a_0 = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{1}{3} \pi^3 \cdot \frac{2}{\pi} = \frac{2}{3} \pi^2$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \cdot dx = \frac{2}{n\pi} \int_0^{\pi} x^2 d\sin nx = \frac{2}{n\pi} \int_0^{\pi} 2x \sin nx dx = \frac{4}{n\pi} \int_0^{\pi} x d\cos nx \\ &= \frac{4}{n^2\pi} (x \cos nx \Big|_0^{\pi} - \int_0^{\pi} \cos nx dx) \\ &= \frac{4}{n^2\pi} (\pi \cdot (-1)^n) = \frac{4}{n^2} (-1)^n \end{aligned}$$

而 $f(x)$ 无间断点, 分片连续且单调

$$\therefore f(x) = 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx + \frac{1}{3} \pi^2$$

$$\begin{aligned} (3) f(x) &= |x|, \quad -\pi \leq x \leq \pi; & (4) f(x) &= \begin{cases} -2, & -\pi \leq x < 0, \\ 1, & 0 \leq x < \pi; \end{cases} \\ (5) f(x) &= \sin^4 x, \quad -\pi \leq x \leq \pi; & (6) f(x) &= \begin{cases} e^x, & -\pi \leq x < 0, \\ 1, & 0 \leq x < \pi. \end{cases} \end{aligned}$$

$$(4) a_0 = \frac{1}{2\pi} \left(\int_{-\pi}^0 -2 dx + \int_0^\pi dx \right) = -1$$

$$\begin{aligned} a_n &= \frac{1}{2\pi} \left(\int_{-\pi}^0 -2 \cos nx dx + \int_0^\pi \cos nx dx \right) \\ &= \frac{3}{2\pi} \int_0^\pi \cos nx dx \\ &= \frac{3}{n2\pi} \sin nx \Big|_0^\pi = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{2\pi} \left(\int_{-\pi}^0 -2 \sin nx dx + \int_0^\pi \sin nx dx \right) \\ &= \frac{3}{2\pi} \int_0^\pi \sin nx dx \\ &= -\frac{3}{n2\pi} \cos nx \Big|_0^\pi \\ &= -\frac{3}{n2\pi} ((-1)^n - 1) \\ &= \frac{3}{n2\pi} [(-1)^{n+1} + 1] \end{aligned}$$

$$\begin{aligned} \therefore f(x) &\sim \sum_{n=1}^{\infty} \frac{3}{n2\pi} [(-1)^{n+1} + 1] \sin nx \\ &= \sum_{n=1}^{\infty} \frac{6}{(2n-1)2\pi} \sin (2n-1)x \\ &= \frac{6}{2\pi} \sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1} \end{aligned}$$

在连续点处分片连续且单周

$$\therefore f(x) = \begin{cases} \frac{6}{2\pi} \sum_{n=1}^{\infty} \frac{\sin (2n-1)x}{2n-1}, & -\frac{1}{2}, x \in (-\pi, \pi) \\ -\frac{1}{2}, & x = \pm\pi. \end{cases}$$

$$(6) a_0 = \frac{1}{2\pi} \left(\int_{-\pi}^0 e^x dx + \int_0^\pi dx \right) = \frac{1}{2\pi} (1 - e^{-\pi} + \pi)$$

$$a_n = \frac{1}{2\pi} \left(\int_{-\pi}^0 e^x \cos nx dx + \int_0^\pi \cos nx dx \right)$$

$$\begin{aligned} \int_0^\pi e^x \cos nx dx &= \int_0^\pi \cos nx de^x = e^x \cos nx \Big|_0^\pi + n \int_0^\pi e^x \sin nx dx \\ &= (-1)^n e^\pi - 1 + n \int_0^\pi e^x \sin nx dx \quad (1) \end{aligned}$$

$$\text{又有 } \int_0^\pi e^x \cos nx dx = \frac{1}{n} \int_0^\pi e^x d \sin nx = \frac{1}{n} e^x \sin nx \Big|_0^\pi - \frac{1}{n} \int_0^\pi e^x \sin nx dx$$

$$\therefore \int_0^\pi e^x \cos nx dx = -\frac{1}{n} \int_0^\pi e^x \sin nx dx \quad (2)$$

$$\therefore \text{联立 (1) (2)} \quad (n^2 + 1) \int_0^\pi e^x \cos nx dx = (-1)^n e^\pi - 1$$

$$\therefore \int_0^\pi e^x \cos nx dx = \frac{(-1)^n e^\pi - 1}{n^2 + 1}$$

$$\therefore a_n = \frac{1}{2\pi} \cdot \frac{(-1)^{n+1} e^\pi + 1}{n^2 + 1}$$

$$b_n = \frac{1}{2\pi} \left(\int_{-\pi}^0 e^x \sin nx dx + \int_0^\pi \sin nx dx \right)$$

$$= \frac{1}{2\pi} \left[\frac{n[(-1)^n e^\pi - 1]}{n^2 + 1} + \frac{1}{n} (-\cos nx) \Big|_0^\pi \right]$$

$$= \frac{1}{2\pi} \frac{n[(-1)^n e^\pi - 1]}{n^2 + 1} + \frac{1}{n2\pi} [(-1)^{n+1} + 1]$$

$$\therefore f(x) \sim \frac{1}{2\pi} (1 - e^{-\pi} + \pi) + \sum_{n=1}^{\infty} \frac{1}{2\pi} \frac{(-1)^{n+1} e^\pi + 1}{n^2 + 1} \cos nx + \left[\frac{1}{2\pi} \frac{n[(-1)^n e^\pi - 1]}{n^2 + 1} + \frac{1}{n2\pi} \right] \sin nx$$

$$= \frac{1}{2\pi} (1 - e^{-\pi} + \pi) + \frac{1}{2\pi} \sum_{n=1}^{\infty} \left[\frac{e^\pi + 1}{(2n+1)^2 + 1} \cos (2n+1)x + \left(\frac{-ne^\pi - 1}{(2n+1)^2 + 1} + \frac{2}{2n+1} \right) \sin (2n+1)x \right]$$

分片连续且单周

2. 将函数 $f(x) = \frac{x^2}{4} - \frac{\pi x}{2}$ ($0 \leq x \leq \pi$) 展开成余弦级数.

3. 将函数 $f(x) = \frac{x^2}{4} - \frac{\pi x}{2}$ ($0 \leq x \leq \pi$) 展开成正弦级数.

$$S(x) = \begin{cases} f(x), & -2 < x < 2 \\ \frac{1+e^{-2}}{2}, & x = \pm 2 \end{cases}$$

2. 将 $f(x)$ 偶延拓,

$$\text{则 } b_n = 0, \quad a_0 = \frac{2}{\pi} \int_0^{\pi} \left(\frac{x^2}{4} - \frac{\pi x}{2} \right) dx = \frac{2}{\pi} \cdot \left(\frac{x^3}{12} - \frac{\pi x^2}{4} \right) \Big|_0^{\pi} = \frac{2}{\pi} \cdot \left(-\frac{\pi^3}{6} \right) = -\frac{\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \left(\frac{x^2}{4} - \frac{\pi x}{2} \right) \cos nx \cdot dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} x^2 \cos nx \cdot dx - \int_0^{\pi} x \cos nx \cdot dx$$

$$\int_0^{\pi} x^2 \cos nx \cdot dx = \frac{1}{n} \int_0^{\pi} 2x \sin nx \cdot dx = \frac{2}{n^2} \int_0^{\pi} x \cdot d \cos nx = \frac{2}{n^2} \cdot \left(x \cos nx \Big|_0^{\pi} - \int_0^{\pi} \cos nx \cdot dx \right) \\ = \frac{2\pi}{n^2} \cdot (-1)^n$$

$$\int_0^{\pi} x \cos nx \cdot dx = \frac{1}{n} \int_0^{\pi} x \cdot d \sin nx = -\frac{1}{n} \int_0^{\pi} \sin nx \cdot dx = \frac{1}{n^2} \cos nx \Big|_0^{\pi} = \frac{(-1)^n - 1}{n^2}$$

$$\therefore a_n = \frac{1}{2\pi} \cdot \frac{2\pi}{n^2} \cdot (-1)^n + \frac{(-1)^{n+1} - 1}{n^2}$$

$$= \frac{(-1)^n}{n^2} - \frac{(-1)^n - 1}{n^2}$$

$$= \frac{1}{n^2}$$

$$\therefore f(x) \sim -\frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$$

3. 将 $f(x)$ 奇延拓, $f(0) = 0$. 无需补充定义

$$a_n = 0, \quad b_n = \frac{2}{\pi} \int_0^{\pi} \left(\frac{x^2}{4} - \frac{\pi x}{2} \right) \sin nx \cdot dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} x^2 \sin nx \cdot dx - \int_0^{\pi} x \sin nx \cdot dx$$

$$\int_0^{\pi} x^2 \sin nx \cdot dx = -\frac{1}{n} \int_0^{\pi} x^2 \cdot d \cos nx = -\frac{1}{n} \left[x^2 \cos nx \Big|_0^{\pi} - \int_0^{\pi} 2x \cos nx \cdot dx \right]$$

$$= \frac{\pi^2}{n} \cdot (-1)^{n+1} + \frac{2}{n} \int_0^{\pi} x \cos nx \cdot dx$$

$$= \frac{\pi^2}{n} \cdot (-1)^{n+1} + \frac{2}{n} \cdot \frac{(-1)^n - 1}{n^2}$$

$$= \frac{2^2}{n} (-1)^{n+1} + \frac{2}{n^3} [(-1)^n - 1]$$

$$\int_0^{2\pi} x \sin nx dx = \frac{2\pi}{n^2} \cdot (-1)^n \cdot \frac{n}{(-2)} = \frac{2}{n} \cdot (-1)^{n+1}$$

$$\begin{aligned} \therefore b_n &= \frac{2}{2n} (-1)^{n+1} + \frac{1}{n^3 \cdot 2} [(-1)^n - 1] - \frac{2}{n} (-1)^{n+1} \\ &= \frac{1}{n^3 \cdot 2} [(-1)^n - 1] + \frac{2}{2n} \cdot (-1)^n \end{aligned}$$

$$\therefore f(x) \sim \left[\frac{2}{2n} (-1)^n + \frac{(-1)^n - 1}{n^3 \cdot 2} \right] \cdot \sin nx$$

4. 求函数

$$f(x) = \begin{cases} \sin \frac{\pi x}{l}, & 0 \leq x < \frac{l}{2}, \\ 0, & \frac{l}{2} \leq x < l \end{cases}$$

的傅里叶正弦级数, 并写出其和函数.

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\frac{\pi}{l}} = 2l.$$

对 $f(x)$ 作奇延拓, $f(0)=0$, 无需补充定义

$$\begin{aligned} a_n &= 0 \\ b_n &= \frac{2}{l} \int_0^{\frac{l}{2}} \sin \frac{2\pi x}{l} \cdot \sin \frac{2\pi}{l} nx \, dx \\ &= \frac{2}{2l} \int_0^{\frac{2\pi}{2}} \sin t \cdot \sin nt \, dt \\ &= \frac{1}{2l} \int_0^{\frac{2\pi}{2}} [\cos(n+1)t - \cos(n-1)t] \, dt \\ &= \frac{1}{2l} \left(\frac{\sin(n+1) \cdot \frac{2\pi}{2}}{n+1} - \frac{\sin(n-1) \cdot \frac{2\pi}{2}}{n-1} \right), \end{aligned}$$

$$n=1 \text{ 时原式} = \frac{2}{2l} \cdot \frac{2\pi}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$n \neq 1 \text{ 时 } \begin{cases} \text{① 当 } n=2k-1 \text{ 时, } b_n=0 \\ \text{② 当 } n=2k \text{ 时, } b_n = \frac{1}{2l} \left(\frac{(-1)^{k-1}}{2k-1} - \frac{(-1)^k}{2k+1} \right) \end{cases}$$

$$= \frac{(-1)^{k-1}}{2} \cdot \frac{4k}{4k^2-1}$$

$$\therefore f(x) \sim \frac{1}{2} \sin \frac{2x}{L} + \sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{2} \frac{4n}{4n^2-1} \sin \frac{2nx}{L}$$

$$S(x) = \begin{cases} \sin \frac{2x}{L}, & 0 \leq x < \frac{L}{2} \\ 1, & x = \frac{L}{2} \\ 0, & \frac{L}{2} \leq x \leq L \end{cases}$$

$k=1$

6. 求函数 $f(x) = \frac{1}{2} - \frac{\pi}{4} \sin x$ ($0 \leq x \leq \pi$) 的傅里叶余弦级数.

7. 求函数 $f(x) = \frac{\pi-x}{2}$ ($0 \leq x \leq 2\pi$) 的傅里叶正弦展开式.

6. 对 $f(x)$ 偶延拓 $b_n = 0, \quad a_0 = \frac{2}{\pi} \int_0^{\pi} \left(\frac{1}{2} - \frac{\pi}{4} \sin x \right) dx = 0$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \left(\frac{1}{2} - \frac{\pi}{4} \sin x \right) \cos nx \cdot dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \cos nx \cdot dx - \frac{1}{2} \int_0^{\pi} \sin x \cos nx \cdot dx$$

$$= -\frac{1}{2} \int_0^{\pi} \sin x \cos nx \cdot dx$$

$$= -\frac{1}{4} \int_0^{\pi} (\sin(n+1)x - \sin(n-1)x) \cdot dx$$

$$= \frac{1}{4} \left(\frac{\cos(n+1)x}{n+1} - \frac{\cos(n-1)x}{n-1} \right) \Big|_0^{\pi}$$

$$= \frac{1}{4} \left(\frac{(-1)^{n+1}}{n+1} - \frac{(-1)^{n-1}}{n-1} - \frac{1}{n+1} + \frac{1}{n-1} \right)$$

$$= \frac{1}{4} \frac{(-1)^{n+1}(n-1) - (-1)^{n-1}(n+1) + 2}{n^2-1}$$

$$= \frac{1}{4} \frac{(-1)^{n-1}(-2) + 2}{n^2-1} = \frac{-2}{4} \frac{(-1)^{n-1}}{n^2-1} = -\frac{1}{2} \frac{(-1)^{n-1}}{n^2-1} = \frac{1}{2} \frac{(-1)^n + 1}{n^2-1}$$

$$\therefore f(x) \sim \sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2-1}$$

$$\therefore \sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2-1} = \frac{1}{2} - \frac{2}{4} \sin x$$

$$7, \quad t=x-\pi, \quad f(t) = \frac{-t}{2}, \quad -\pi < t < \pi$$

$f(t)$ 为奇函数, $f(0)=0$.

$$\begin{aligned} a_n &= 0, \quad b_n = -\frac{1}{2} \cdot \frac{2}{\pi} \cdot \int_0^{\pi} t \sin nt \, dt \\ &= \frac{1}{n\pi} \cdot \int_0^{\pi} t \, d\cos nt = \frac{1}{n\pi} \cdot (t \cos nt \Big|_0^{\pi} - \int_0^{\pi} \cos nt \, dt) \\ &= \frac{1}{n\pi} \cdot \pi \cdot (-1)^n = \frac{(-1)^n}{n} \end{aligned}$$

$$\therefore f(t) \sim \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \cdot \sin nt.$$

$$\begin{aligned} f(x) &\sim \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \cdot \sin n(x-\pi) \\ &= \sum_{n=1}^{\infty} \frac{\sin nx}{n}, \quad x \in (0, 2\pi) \end{aligned}$$

$$1. \quad x=0/2\pi \text{ 时, } S(x)=0$$

8. 利用前面各题中的展开式, 求下列级数的值.

$$(1) \sum_{n=1}^{\infty} \frac{\sin n}{n}; \quad (2) \frac{1}{2^2} + \frac{1}{4^2} + \cdots + \frac{1}{(2n)^2} + \cdots$$

$$(1) \text{ 由 7 知 } \frac{2-x}{2} = \sum_{n=1}^{\infty} \frac{\sin nx}{n}, \quad \text{令 } x=1, \text{ 得 } \sum_{n=1}^{\infty} \frac{\sin n}{n} = \frac{2-1}{2}$$

$$(2) \text{ 由 2 知 } \frac{x^2}{4} - \frac{2x}{2} = \sum_{n=1}^{\infty} \frac{\cos nx}{n^2} - \frac{2^2}{6}, \quad \text{令 } x=0, \text{ 有 } \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{2^2}{6}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{4n^2} = \frac{2^2}{24}$$

9. 设函数 $f(x)$ 以 T 为周期, 它在一个周期内的表达式为

$$f(t) = \begin{cases} 0, & -T/2 \leq t < 0, \\ A \sin \omega t, & 0 \leq t < T/2, \end{cases}$$

其中 $\omega = \frac{2\pi}{T}$, $A > 0$, 求 $f(t)$ 的傅里叶展开式.

$$\begin{aligned} 9. \quad a_0 &= \frac{2}{T} \int_0^{\frac{T}{2}} A \sin \omega t \cdot dt = \frac{-2A}{T\omega} \cdot \cos \omega t \Big|_0^{\frac{T}{2}} \\ &= \frac{-2A}{T\omega} \left(\cos \frac{2\omega}{T} \cdot \frac{T}{2} - 1 \right) \\ &= \frac{-A}{\omega} (-2) = \frac{2A}{\omega} \end{aligned}$$

$$\begin{aligned} a_n &= \frac{2}{T} \int_0^{\frac{T}{2}} A \sin \frac{2\omega}{T} t \cos \frac{2\omega}{T} n t \cdot dt \\ &\quad \text{令 } \frac{2\omega}{T} t = x, \therefore t = \frac{T}{2\omega} x \\ \therefore a_n &= \frac{A}{\omega} \int_0^{\omega} \sin x \cdot \cos n x \cdot dx \\ &= \frac{A}{\omega} \int_0^{\omega} [\sin(n+1)x - \sin(n-1)x] dx \\ &= \frac{A}{\omega} \left(\frac{\cos(n+1)x}{n+1} - \frac{\cos(n-1)x}{n-1} \right) \Big|_0^{\omega} \\ &= \frac{A}{\omega} \left(\frac{(-1)^{n+1}}{n+1} - \frac{(-1)^{n-1}}{n-1} - \frac{1}{n+1} + \frac{1}{n-1} \right) \\ &= \frac{A}{\omega} \frac{(n+1)(-1)^{n+1} - (n-1)(-1)^{n-1} - 2}{n^2-1} \\ &= \frac{A}{\omega} \frac{(-1)^{n+1} \cdot 2 - 2}{n^2-1} = \frac{A}{\omega} \frac{(-1)^{n+1} - 1}{n^2-1} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^{\frac{T}{2}} A \sin \frac{2\omega}{T} t \cdot \sin \frac{2\omega}{T} n t \cdot dt \\ &\quad \text{令 } \frac{2\omega}{T} t = x, \text{ 则} \\ b_n &= \frac{A}{\omega} \int_0^{\omega} \sin x \cdot \sin n x \cdot dx, \quad n \neq 1 \text{ 时} \\ &= -\frac{A}{\omega} \int_0^{\omega} [\cos(n+1)x - \cos(n-1)x] dx \\ &= -\frac{A}{\omega} \left(\frac{\sin(n+1)x}{n+1} - \frac{\sin(n-1)x}{n-1} \right) \Big|_0^{\omega} \\ &= 0 \\ n=1 \text{ 时, } b_1 &= \frac{A}{\omega} \cdot 2 \cdot \frac{\omega}{4} = \frac{A}{2} \end{aligned}$$

$$\begin{aligned} \therefore f(t) &\sim \frac{A}{\omega} + \sum_{n=1}^{\infty} \frac{A}{\omega} \frac{(-1)^{n+1} - 1}{n^2-1} \cos \frac{2\omega}{T} n x \\ &= \frac{A}{\omega} + \frac{A}{\omega} \sum_{n=1}^{\infty} \frac{-2}{4n^2-1} \cos \frac{2\omega}{T} 2n x \\ &= \frac{A}{\omega} - \frac{2A}{\omega} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos \frac{4n\omega}{T} x + \frac{A}{2} \sin \frac{2\omega}{T} x \\ &= \frac{A}{\omega} + \frac{A}{2} \sin \omega x - \frac{2A}{\omega} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos 2\omega n x \end{aligned}$$

10. 设函数 $f(x)$ ($-\pi \leq x \leq \pi$) 的傅里叶系数为 a_0, a_n, b_n ($n=1, 2, \dots$), 求函数 $g(x) = f(-x)$ ($-\pi \leq x \leq \pi$) 的傅里叶系数 A_0, A_n, B_n ($n=1, 2, \dots$).

11. 设函数 $f(x)$ 是以 2π 为周期的连续函数, a_0, a_n, b_n ($n=1, 2, \dots$) 为其傅里叶系数, 求函数

$$F(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) f(t+x) dt$$

的傅里叶系数 A_0, A_n, B_n ($n=1, 2, \dots$).

$$a_0 = \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(x) dx, \quad A_0 = \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(-x) dx = \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(x) dx = a_0$$

$$\begin{aligned} a_n &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(x) \cos nx dx, & A_n &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(-x) \cos nx dx \\ & & &= -\frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(-x) \cos n(-x) d(-x) \\ & & &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(x) \cos nx dx = a_n \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(x) \sin nx dx, & B_n &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(-x) \sin(-nx) d(-x) \\ & & &= -b_n \end{aligned}$$

$$\begin{aligned} 11. \quad A_0 &= \frac{1}{2\pi} \int_{-2\pi}^{2\pi} F(x) dx = \frac{1}{2\pi} \int_{-2\pi}^{2\pi} dx \int_{-2\pi}^{2\pi} f(t) f(t+x) dt \\ &= \frac{1}{2\pi^2} \int_{-2\pi}^{2\pi} dt \int_{-2\pi}^{2\pi} f(t) f(t+x) dx \\ &= \frac{1}{2\pi^2} \int_{-2\pi}^{2\pi} f(t) dt \int_{-2\pi+t}^{2\pi+t} f(s) ds \end{aligned}$$

$$\text{由于 } f(x) \text{ 以 } 2\pi \text{ 为周期, 则 } \int_{-2\pi+t}^{2\pi+t} f(s) ds = \int_{-2\pi}^{2\pi} f(s) ds = 2\pi a_0$$

$$\therefore A_0 = \frac{1}{2\pi^2} \int_{-2\pi}^{2\pi} 2\pi a_0 f(t) dt = a_0^2$$

$$A_n = \frac{1}{2\pi} \int_{-2\pi}^{2\pi} F(x) \cos nx dx = \frac{1}{2\pi^2} \int_{-2\pi}^{2\pi} \cos nx dx \int_{-2\pi}^{2\pi} f(t) f(t+x) dt$$

$$a_n = \frac{1}{2\pi} \int_{-2\pi}^{2\pi} f(s) \cos ns ds$$

$$= \frac{1}{2} \cdot \int_{-2}^2 f(t) dt \cdot \int_{-2}^2 f(t+x) \cos nx \cdot dx$$

$$\int_{-2}^2 f(t+x) \cos nx dx = \int_{-2+t}^{2+t} f(s) \cos n(s-t) ds$$

$$= \int_{-2}^2 f(s) [\cos ns \cos nt + \sin ns \sin nt] ds$$

$$= 2(\cos nt \cdot a_n + \sin nt \cdot b_n)$$

$$\therefore A_n = \frac{1}{2} \int_{-2}^2 f(t) \cos nt \cdot a_n + f(t) \sin nt \cdot b_n dt$$

$$= a_n^2 + b_n^2$$

$$\therefore F(-x) = \frac{1}{2} \int_{-2}^2 f(t) \cdot f(t-x) dt, \text{ 令 } t-x = s$$

$$\therefore F(-x) = \frac{1}{2} \int_{-2-x}^{2-x} f(s+x) \cdot f(s) ds = \frac{1}{2} \int_{-2}^2 f(x+t) \cdot f(t) dt = F(x)$$

$\therefore F(x)$ 为偶函数, 有

$$B_n = 0$$

习 题 12.3

利用帕塞瓦尔等式,求下列级数的和:

$$1. \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} \quad (\text{提示: 利用函数 } f(x)=|x| \quad (-\pi \leq x \leq \pi) \text{ 的傅里叶系数}).$$

$$2. \sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^2} \quad (0 < \varphi < \pi) \quad (\text{提示: 利用函数}$$

$0 \leq x \leq \varphi$).

$$f(x) = \begin{cases} 1, & 0 \leq x \leq \varphi, \\ -1, & -\varphi \leq x < 0, \\ 0, & \varphi < |x| \leq \pi \end{cases}$$

的傅里叶系数).

Paseval 等式: $\frac{a_0^2}{2} + \sum_{k=1}^{\infty} (a_k^2 + b_k^2) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)^2 dx$

$$2. f(x) = \begin{cases} 1, & 0 \leq x \leq \varphi \\ -1, & -\varphi \leq x < 0 \\ 0, & \varphi < |x| \leq \pi \end{cases}$$

(禁止: 负号不能放在积分限上!)

$$a_0 = \frac{1}{2\pi} \int_{-\varphi}^0 -1 dx + \frac{1}{2\pi} \int_0^{\varphi} 1 dx$$

$$= \frac{1}{2\pi} \left(-x \Big|_{-\varphi}^0 + x \Big|_0^{\varphi} \right) = 0$$

$$a_n = \frac{1}{2\pi} \int_0^{\varphi} \cos nx \cdot dx = 0$$

$$b_n = \frac{2}{2\pi} \int_0^{\varphi} \sin nx \cdot dx = \frac{-2}{2\pi} \cdot \cos nx \Big|_0^{\varphi} = \frac{2}{\pi n} (1 - \cos n\varphi)$$

$$\therefore \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} (1 - 2\cos n\varphi + \cos^2 n\varphi) = \frac{1}{2\pi} \int_{-\varphi}^{\varphi} f(x)^2 dx$$

$$\therefore \frac{4}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{2\cos n\varphi}{n^2} + \frac{\cos^2 n\varphi}{n^2} \right) = \frac{2\varphi}{\pi}$$

$$\therefore \frac{2}{\pi} - \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^2} + \frac{4}{\pi^2} \left(\frac{\pi^2}{6} - \frac{2\varphi}{2} + \frac{\varphi^2}{2} \right) = \frac{2\varphi}{\pi}$$

$$\therefore \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos n\varphi}{n^2} = \frac{2}{3} + \frac{2}{3} - \frac{2\varphi}{\pi} + \frac{2\varphi^2}{\pi^2} - \frac{2\varphi}{\pi}$$

$$= \frac{2}{3} + \frac{2\varphi^2}{\pi^2} - \frac{4\varphi}{\pi}$$

$$a_n = \frac{2}{\pi n^2} [(-1)^n - 1]$$

$$b_n = 0$$

$$\frac{\pi^2}{2} + \sum_{n=1}^{\infty} \frac{4}{\pi^2 n^4} [(-1)^n - 1]^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2\pi^3}{32} = \frac{2}{3}\pi^2$$

$$\frac{\pi^2}{2} + \sum_{n=1}^{\infty} \frac{16}{\pi^2 (2n-1)^4} = \frac{2}{3}\pi^2$$

$$\therefore \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{1}{6}\pi^2$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

习 题 12.3

1. $\frac{\pi^4}{96}$. 2. $\left(\frac{2\pi^2 - 6\pi\varphi + 3\varphi^2}{12}\right)$.

$$\begin{aligned}\therefore \sum_{n=1}^{\infty} \frac{\ln n^4}{n^2} &= \frac{2^2}{8} \left(\frac{4}{3} + \frac{2\varphi^2}{2^2} - \frac{4\varphi}{2} \right) \\ &= \frac{2^2}{6} + \frac{\varphi^2}{4} - \frac{2\varphi}{2}\end{aligned}$$